DataBase System Lab #02

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Class Tasks:

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DataBase System Lab 2

Activity #01:

1. Restaurant Management System

Customers → Menu Items → Orders → Chefs → bills.

2. Airport System

Passengers → Flights → boarding Passes → Airline Staff → gates.

Activity #02:

| Library System. | ~~Justification.~~ |
|---|---|
| Entity / Not Entity | Justification. |
| Book (Entity) | Core Object: such as title, author, ISBN, genre |
| Member (Entity) | Person using library services; information like name, ID, contact. |
| Librarian (Entity) | Employee managing library operations; employeeID, name, department |
| Borrowing Record (Entity) | Tracks book issue & return information, including dates & over due |
| Reservation (Entity) | Stores information about reserved books. |
| Book request (Entity) | Recording member requests for new books. |
| Replacement Record (Entity) | Tracks books that are replaced/repaired to maintain inventory. |
| Event (Entity) | Represents book clubs, guest lectures |
| Library Building (Not Entity) | Physical structures, no operational data is stored is db. |
| Reading hall (Not Entity) | Facility for reading, not relevant to database records. |
| Coffee Corner (Not Entity) | Convenience area; unrelated to library |
| Computer Stations (Not Enti) | Equipment for catelog access, not tracked as entities. |
| Cleaning staff (Not Entity) | Support staff, not part of library information system. |

Signature

Home Tasks:

24K-0725 | Muhammad Adil Saeed.

Scenario 1: Community health Clinic System.

Step I: Identification of Mini-world.
The mini-world is the community health clinic management system, which manages data related to medical camps, medical supplies, patients and clinic staff. Facilities like the waiting areas, pharmacy and parking are excluded since no data is stored.

Step II: Identification of Entities

| Enitity | Reason |
|---|---|
| Medical camp | Real-world activity with multiple details such as name, service, schedule and fee. |
| Medical Supply | Physical item used in camps with important details like quantity and cost. |
| Patient | Individual registered at clinic. |
| Staff | Doctors and staff members working |

Step III: Identification of Attributes.

| Enitity | Attributes |
|---|---|
| Medical Camp | CampName, HealthService, Schedule, Registration Fee. |
| Medical Supply | ItemName, Quantity, Cost. |
| Patient | PatientID, Name, Age, ContactDetails. |
| Staff | EmployeeID, Name, Specialization. |

Step IV: Assignment of Data.

| Entity | Attributes | Data Type | Reason |
|---|---|---|---|
| Medical Camp | Camp Name | Text/String | Contains words. |
| | Health Services | Text/String | For description of service. |
| | Schedule | Text/String | Store date/time info |
| | Registration fee | Decimal/Float | Fee may contain decimals. |
| Medical Supply | Item Name | Text/String | Contain words. |
| | Quantity | Number/Integer | Quantity in numeric. |
| | Cost. | Decimal/Float | Cost may in decimals. |
| Patient | Patient ID | Number/Integer. | Unique identifier for patient |
| | Name | Text/String | Contain letters. |
| | Age | Number/Integer | Age is numeric |
| | Contact Details | String/Text. | Stores phone/email info |
| Staff. | Employee ID. | Number/Integer | Unique identifier for staff |
| | Name | Text/String | Contains letters & words. |
| | Specialization. | Text/String | Describe expertise. |

Final Note:
Facilites such as the waiting areas, pharmacy and parking space are not considered entities because the system does not store any related data to them.

Scenario 2: Language Learning Institute System.

Step I: Identification of Mini World.

The mini world is the Language Learning Institute Management System, which manages data related to courses, learning materials, students and instructors. Facilities like the library, student lounge and reception area are excluded.

Step II: Identification of Entities.

| Entity | Reason |
|---|---|
| Courses. | Academic Offering with details such as title, language, duration and fee. |
| Learning material | Educational resources like books and audio materials with quantity and cost. |
| Student. | Individual enrolled in courses. |
| Instructor | People responsible for teaching courses. |

Step III: Identification of Attributes.

| Entity | Attributes. |
|---|---|
| Courses | CourseTitle, LanguageType, Duration, Fee. |
| Learning material | MaterialName, Quantity, Cost. |
| Student. | StudentID, Name, ProficiencyLevel. |
| Instructor | EmployeeID, Name, Teaching Specialization. |

Step IV : Assignment of Data Types.

| Entity | Attribute | Data Type | Reason |
|---|---|---|---|
| Cource | Course Title | Text/String | Contains words |
| | Language Type | Text/String | Describe Language |
| | Duration | Number/Integer | Duration in months/hours |
| | Fee | Decimal / Float | Fee may be in decimals. |
| Learning Material | Material Name | Text/String | Contain words |
| | Quantity | Number/Integer | Quantity is numeric |
| | Cost | Decimal / Float | Cost may include decimals. |
| Student | Student ID | Number/Integer | ~~Contains~~ Unique identifier. |
| | Name | Text/String | Contains letters & words. |
| | Proficiency level. | Text/String | Describes skill level. |
| Instructor | Employee ID | Number/Integer | Unique identifier. |
| | Name | Text/String | contains letters & words. |
| | Teaching Specialization | Text/String | Describe area of teaching |

Final Note

Facilities such as library, student lounge & reception area are not treated as entities because the system does not store or manage any information about them.